

# The Garg–Pagano benchmark from zero

Problem statement, statics, MSG-RM chain, FSDT–Whitney chain:  
every formulation and the code that generated every number

Repository: `OpenSG-TW`, branch `rm-xsec-dehom`, folder `examples/garg/`. All equation numbers cite: Garg et al., Compos. Struct. 305 (2023) 116551; Whitney, J. Appl. Mech. 40 (1973) 302–304; Yu, Hodges & Volovoi, Comput. Struct. 81 (2003) 439–454; Pagano, J. Compos. Mater. 3 (1969) 398–411.

## 1 Step 0 — the problem statement (Pagano / Garg)

A laminated strip, span  $0 \leq x \leq a$  ( $a = 1$  m), thickness  $-h/2 \leq z \leq h/2$ , infinite across the width (cylindrical bending: nothing varies with  $y$ ,  $\varepsilon_{22} = 0$ ), simply supported, carrying a sinusoidal pressure on the top face only:

$$\sigma_{33}(x, +\frac{h}{2}) = q_0 \sin(px), \quad p = \pi/a, \quad q_0 = 10^4 \text{ Pa} \quad (1)$$

$$\sigma_{33}(x, -\frac{h}{2}) = \sigma_{13}(x, \pm\frac{h}{2}) = 0, \quad w = \sigma_{11} = 0 \text{ at } x = 0, a \quad (\text{Garg Eq. 21})$$

with continuity of  $(u, w, \sigma_{13}, \sigma_{33})$  at every ply interface (Garg Eq. 22).

**The three laminate families** (`garg_layups.py`; total thickness  $h = a/S$ ):

| case | stack                        | material                                                                            | aspect ratios $S = a/h$ |
|------|------------------------------|-------------------------------------------------------------------------------------|-------------------------|
| A    | [0/90/0], equal thirds       | Pagano: $E_1=25, E_2=E_3=1$ GPa,<br>$G_{12}=G_{13}=0.5, G_{23}=0.2$ GPa, $\nu=0.25$ | 10, 50                  |
| B    | [0/90/90/0], quarters        | AS4: $E_1=181, E_2=10.3$ GPa                                                        | 10, 50                  |
| C    | [0/core/0], $0.1h/0.8h/0.1h$ | faces $E_1=131$ GPa; core $E=0.578$ GPa                                             | 10, 50                  |

**Harmonic families and stations.** Every field is one harmonic:  $\sigma_{11}, \sigma_{22}, \sigma_{33}, w \sim \sin(px)$  (peak at  $x = a/2$ );  $\sigma_{13}, u \sim \cos(px)$  (peak at  $x = 0$ ). Hence the through-thickness comparisons:  $\sigma_{13}$  at  $x = 0$ ,  $\sigma_{33}$  at  $x = a/2$  — the same stations Garg reports.

**Reference surface:** the laminate *mid-surface* in every chain (SG `fraction=0.5`; the exact solver’s  $z$  origin).

## 2 Step 1 — statics of the resultants (shared input; NO material property)

Sectional equilibrium of a slice  $[x, x + dx]$ :

$$\frac{dQ_1}{dx} = -q(x), \quad \frac{dM_{11}}{dx} = Q_1(x) \quad (2)$$

Integrating and applying the simple-support conditions  $M_{11}(0) = M_{11}(a) = 0$  (which force both integration constants to zero since  $\sin \pi = 0$ ):

$$Q_1(x) = \frac{q_0}{p} \cos(px), \quad M_{11}(x) = \frac{q_0}{p^2} \sin(px) \quad (3)$$

At the  $\sigma_{13}$  station  $x = 0$ :  $Q_1 = q_0/p = 3183.10$  N/m (equal to the support reaction),  $M_{11} = 0$ . The strip is statically determinate — this same  $Q_1$  is what an Abaqus plate solve returns (SF4

= 3182.44, measured) and what integrating the exact stresses gives (3183.04): one statics value, three routes.

*Code:* `statics_fsdt.py :: statics_resultants(q0, a, x)` — two lines implementing the boxed result. Called in `pagano_bench.py::run_case`:

```
Q1, _ = statics_resultants(q0, a, 0.0)      # x = 0 : Q1 = q0/p
FF_end = np.array([0, 0, 0, 0, 0, 0, Q1, 0])
```

### 3 Step 2 — Chain I: MSG-RM

#### 2a. The 1-D structure gene from the layup

The plate SG is the wall discretised through its thickness: 5-noded (quartic) elements, one per ply, nodes from  $-h/2$  to  $+h/2$  (mid-surface reference). Quartic is the degree that represents the whole warping ladder exactly per ply ( $V_0$  quadratic,  $V_1$  cubic,  $V_2$  quartic). *Code:* `opensg-jax/fe-jax/segment_plate.py` (mesh generator); layups defined in `examples/garg/garg_layups.py`.

#### 2b. Homogenization: the $8 \times 8$ plate law

One call per laminate:

```
r = rm_plate_msg(thk, ang, mats, MATERIAL_DB, fraction=0.5)
```

*Code:* `opensg-jax/fe-jax/msg_rm_plate.py`. Internally (equation numbers = Yu 2003): assemble the SG matrices  $E = \langle [\Gamma_h S]^T C [\Gamma_h S] \rangle$ ,  $D_{h\epsilon}$ ,  $D_{hl_a}$ ,  $D_{la\epsilon}$ ,  $D_{la\epsilon}$  (Eq. 30); solve the constrained zeroth-order warping  $EV_0 + D_{h\epsilon}\epsilon = H\psi\Lambda_0$  (Eq. 34, KKT) giving the classical stiffness  $A = D_{\epsilon\epsilon} + \hat{V}_0^T D_{h\epsilon}$  (Eq. 40); solve the twelve first-order columns  $EV_{1a} + \bar{D}_a = H\psi(\psi^T \bar{D}_a)$  with  $\bar{D}_a = (D_{hl_a} - D_{hl_a}^T)\hat{V}_0 - D_{la\epsilon}$  (Eqs. 43–45); project the second-order energy onto the Reissner–Mindlin form by least squares over  $X = G^{-1}$  and Yu’s 24 relaxed constants (Eqs. 56–61)  $\rightarrow$  the transverse shear stiffness  $G$ . Returned:  $ABDG = [[A, B, 0], [B, D, 0], [0, 0, G]]$  ( $8 \times 8$ ) plus the warping fields  $V_0, V_{11}, V_{12}$  used by the recovery.

#### 2c. Resultants $\rightarrow$ plate strains (constitutive + one equilibrium step)

$$\epsilon = A_6^{-1} \text{FF}[6:] \quad (\text{here } \epsilon = 0: \text{ only } Q_1 \text{ acts at } x=0) \quad (4)$$

$$\boxed{\epsilon_{,1} = A_6^{-1} [0, 0, 0, Q_1, 0, 0]^T} \quad (\text{equilibrium } M_{11,1} = Q_1) \quad (5)$$

The closure was validated against the oracle gradient taken from the exact solution:  $\sigma_{13}$ -profile difference 0.008% ( $S=10$ ) / 0.02% ( $S=4$ ).

```
S6      = np.linalg.inv(np.asarray(r["A6"]))
E6_end  = S6 @ FF_end[:6]
dE1_end = S6 @ np.array([0, 0, 0, Q1, 0, 0.0])
```

## 2d. Pointwise recovery (Yu Eqs. 63/66) and $\sigma_{33}$

The implemented recovery is the full SECOND-order formula (Eq. 66), carrying the  $V_2$  warping blocks of Eq. 64 ( $\bar{V}_{1a} = V_{1a} + \text{kernel } c_a$  are the relaxed first-order columns of Eq. 58):

$$\Gamma(z) = \Gamma_h S(V_0 \epsilon + \bar{V}_{11} \epsilon_{,1} + \bar{V}_{12} \epsilon_{,2} + V_{21} \epsilon_{,11} + V_{22} \epsilon_{,12} + V_{23} \epsilon_{,22}) + \Gamma_\epsilon \epsilon \\ + \Gamma_{l_1} S(V_0 \epsilon + \bar{V}_{11} \epsilon_{,1} + \bar{V}_{12} \epsilon_{,2})_{,1} + \Gamma_{l_2} S(V_0 \epsilon + \bar{V}_{11} \epsilon_{,1} + \bar{V}_{12} \epsilon_{,2})_{,2}, \quad \sigma(z) = C(z) \Gamma(z) \quad (6)$$

For THIS load case the  $V_2$  terms carry exactly zero, for two independent reasons: (i) the constant- $Q$  statics state has  $\epsilon_{,11} = \epsilon_{,12} = \epsilon_{,22} = 0$  (the second gradients of the resultants vanish when  $Q$  is constant), and (ii) for monoclinic laminates the  $V_2$  contribution to  $\sigma_{13}, \sigma_{23}$  is identically zero by the parity ladder regardless of the load ( $V_2$ 's payload is  $\sigma_{33}$ , which enters here through the equilibrium integration below). The code nevertheless evaluates the full formula (`msgrm_strain_at_depth` accepts `dE11/dE12/dE22`). The  $\sigma_{13}$  content is carried by  $\bar{V}_{11} \epsilon_{,1}$ ; the traction-free faces emerge naturally.  $\sigma_{33}$  by through-thickness equilibrium of the recovered amplitude (families  $\sigma_{13} \sim \cos \Rightarrow \sigma_{33} \sim \sin$ ):

$$\hat{\sigma}_{33}(z) = \int_{-h/2}^z p \hat{\sigma}_{13}(z') dz' \quad (\text{lands on } q_0 \text{ at the top face in every case}) \quad (7)$$

$\sigma_{33}$ : *equilibrium integration, and its relation to Yu's direct route.* Yu's 2003 recovery paper obtains  $\sigma_{33}$  from the 3-D constitutive law applied to the recovered strain ("it is not necessary to use integration through the thickness of the 3-D equilibrium equations ..."), which works because his implementation carries the LOAD COLUMNS forced by the surface traction (his Eqs. 29 and 45:  $V_1 = V_{11} \epsilon_{,1} + V_{12} \epsilon_{,2} + V_{1L}$ ). Those columns were prototyped in the present core and validated on this benchmark (faces exact by construction;  $\sigma_{33}$  1.19% at  $S=10$ , case A) and then REMOVED by decision: the through-thickness equilibrium integration used here,  $\hat{\sigma}_{33}(z) = \int p \hat{\sigma}_{13} dz'$  (the pointwise 3-D equilibrium  $\sigma_{13,1} + \sigma_{33,3} = 0$  applied to the recovered  $\sigma_{13}$  — no assumption), delivers equal or better accuracy for the pressure-loaded cases without extending the recovery interface, and for beam-resultant loading (the code's primary use) the load drivers vanish identically. The two routes were measured asymptotically equivalent ( $\sim 0.1$ -point agreement on the laminates); the working load-column implementation is preserved in the repository history (commits `d1b52ac/3e2f1d0`).

for i, z in enumerate(zc):

    s13\_m[i] = msgrm\_strain\_at\_depth(r, z, E6\_end, dE1\_end, z6)[1][4]

s33\_m = cumtrapz(p \* s13\_m, zc)      # equilibrium integration

*Code:* recovery in `msg_rm_plate.py::msgrm_strain_at_depth`; driver loop in `pagano_bench.py::run_case`.

## 4 Step 3 — Chain II: FSDT with the Whitney-1973 correction

Input: the same statics  $Q_1 = q_0/p$ . Whitney's construction (his equation numbers):

**(Eq. 1) The FSDT constitutive law.**  $Q_1 = k_1^2 A_{55} \gamma$  with  $A_{55} = \int C_{55}(z) dz$  and  $\gamma$  constant through the thickness (the FSDT kinematic assumption).

**(Eq. 3) The bending-stress gradient** under constant shear (classical lamination;  $\bar{Q}_{11}(z) =$  plane-stress reduced ply stiffness):

$$\frac{\partial \sigma_{11}}{\partial x}(z) = \bar{Q}_{11}(z) \frac{B_{11} - A_{11}z}{D} Q_1, \quad (A_{11}, B_{11}, D_{11}) = \int \bar{Q}_{11}(1, z, z^2) dz, \quad D = A_{11}D_{11} - B_{11}^2 \quad (8)$$

**(Eqs. 4–5) The equilibrium shear shape.** Feeding Eq. (3) into  $\sigma_{11,1} + \sigma_{13,3} = 0$  and integrating from the free bottom face:

$$\sigma_{13}(z) = \hat{g}(z) Q_1, \quad \hat{g}(z) = \frac{1}{D} \int_{-h/2}^z \bar{Q}_{11} (B_{11} - A_{11} z') dz' \quad (9)$$

Provable and *asserted in the code*:  $\hat{g}(\pm h/2) = 0$  (both faces free) and  $\int \hat{g} dz = 1$  (the shape carries exactly  $Q_1$ ; by parts,  $(A_{11} D_{11} - B_{11}^2)/D = 1$ ).

**(Eq. 7) The correction factor** — equate the FSDT shear energy to the complementary energy of that shape:

$$\frac{Q_1^2}{2k_1^2 A_{55}} = \frac{1}{2} \int \frac{[\hat{g}(z) Q_1]^2}{C_{55}(z)} dz \quad \Rightarrow \quad \boxed{k_1^2 = \left( A_{55} \int \hat{g}^2 / C_{55} dz \right)^{-1}} \quad (10)$$

Self-checks in the module: homogeneous  $\rightarrow k_1^2 = 5/6$  exactly (Reissner); [0/90/0] Pagano material  $\rightarrow 0.5828$ , the regime of Whitney’s own 4-ply example (0.5952).

**The standalone FSDT outputs** (what the  $k$  is then used for):

$$\sigma_{13}(z) = C_{55}(z) \frac{Q_1}{k_1^2 A_{55}} \quad (\text{layerwise-constant staircase}), \quad \sigma_{33} \equiv 0, \quad w\left(\frac{a}{2}\right) = \frac{q_0}{p^4 D_{11}} + \frac{q_0}{p^2 k_1^2 A_{55}} \quad (11)$$

Whitney uses the good shape  $\hat{g}$  *only* to calibrate the scalar  $k$ ; the shape is then discarded — which is why no  $k$  can repair the staircase profile, and why his own validation (his Fig. 1) is deflection-only.

*Code*: `statics_fsdt.py` — `whitney_k1sq` (exact per-ply moments  $A_{11}, B_{11}, D_{11}$ ; cumulative integral for  $\hat{g}$ ; the two assertions; Eq. 7) and `fsdt_s13` (the staircase). FSDT does not touch Abaqus, Pagano, or the MSG core.

## 5 Step 4 — the reference: Pagano’s exact solution (curves only)

The per-ply 3-D equations reduce, for one harmonic, to a  $6 \times 6$  state-space system in  $z$  with state  $[u, v, w, \sigma_{13}, \sigma_{23}, \sigma_{33}]$ , propagated by transfer matrices with interface continuity and the face tractions of the problem statement imposed exactly (boundary residual  $\sim 10^{-15}$ ; resultant closure exact). Equivalent to Pagano’s  $f_i(y)$  construction (Garg Eqs. 23–24). *Code*: `pagano_exact.py` (entry point, families documented)  $\rightarrow$  `examples/TW-paper/rm_thickness/exact_cyl.py` (the solver). The exact solution appears *only* as the reference curves — never as an input.

## 6 Step 5 — the data map and results

| file                                                      | role                                        | outputs                          |
|-----------------------------------------------------------|---------------------------------------------|----------------------------------|
| <code>garg_layups.py</code>                               | the three layups + material db              | <code>case*/garg*_sg.yaml</code> |
| <code>pagano_exact.py</code>                              | exact reference                             | (curves in memory)               |
| <code>statics_fsdt.py</code>                              | statics $Q_1$ + Whitney $k_1^2$ + staircase | (values in memory)               |
| <code>pagano_bench.py</code>                              | orchestrator (MSG chain + plots + tables)   | <code>pagano_S*.dat, .png</code> |
| <code>case*/7_helper_RM_Pagano_benchmark{1,2,3}.py</code> | per-case drivers                            | run everything above             |

Each `pagano_S⟨S⟩.dat` holds: the RM  $8 \times 8$ , the statics FF, both shear-stiffness constructions ( $G_{\text{Whitney}}, G_{\text{MSG}}, k_1^2$ ), the error summary, and the pointwise table  $z, \sigma_{13}^{\text{MSG}}, \sigma_{13}^{\text{exact}}, \sigma_{13}^{\text{FSDT}}, \sigma_{33}^{\text{MSG}}, \sigma_{33}^{\text{exact}}$ .

**Relative  $L_2$  errors vs the exact solution** ( $\sigma_{33}$  MSG = the DIRECT constitutive recovery via the load columns; the equilibrium-integration cross-check in parentheses):

| case / $S$ | $\sigma_{13}$ MSG | $\sigma_{13}$ FSDT | $\sigma_{33}$ MSG |
|------------|-------------------|--------------------|-------------------|
| A 10       | 4.43 %            | 107.1 %            | 1.16 %            |
| A 50       | 0.18 %            | 108.7 %            | 0.048 %           |
| B 10       | 1.19 %            | 91.9 %             | 0.31 %            |
| B 50       | 0.041 %           | 92.4 %             | 0.010 %           |
| C 10       | 9.17 %            | 5269 %             | 0.82 %            |
| C 50       | 0.39 %            | 5468 %             | 0.034 %           |

( $\sigma_{33}$  FSDT does not exist. The aspect ratios are restricted to the PLATE-MODEL REGIME —  $S = 10$  (moderately thick) and  $S = 50$  (thin) — since at  $S \leq 5$  the first-order plate truncation itself dominates every plate theory and the comparison stops discriminating between them. The FSDT numbers use the strict Whitney-1973  $k_1^2$ ; the energy-correct  $k$  *worsens* the pointwise staircase relative to the naive 5/6 —  $k$  repairs deflections, never shapes.)

**Reproduce everything:**

```
python examples/garg/garg_layups.py           # the three SGs
python examples/garg/caseA/7_helper_RM_Pagano_benchmark1.py # A: S = 4, 10, 100
python examples/garg/caseB/7_helper_RM_Pagano_benchmark2.py # B: S = 5, 10, 100
python examples/garg/caseC/7_helper_RM_Pagano_benchmark3.py # C: S = 4, 10, 100
```